Design and Development of a Swing Motion-Based Electrical Power Generator



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This research project report was submitted to the Department of Engineering at the University of Zululand in partial fulfilment of the academic requirements for a Bachelor of Engineering degree in Electrical Engineering.

Key words: Kinetic energy harvesting, swing-powered harvesting, renewable energy.

Declaration

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LECTURER REMARKS		

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Abstract

This project presents the design, simulation and construction of a swing-powered electrical generator that converts the kinetic energy of the back-and-forth motion of a swing into usable electrical energy. The aim of the project was to investigate a small-scale method of renewable energy generation that uses an everyday source of motion. The system uses a bicycle freewheel to convert the oscillating motion of the swing into one-directional rotation, a chain drive to increase the speed, and a permanent magnet DC generator to produce the electrical output. A DC boost module was used to raise the low generator voltage to a usable level.

The swing was modelled as a nonlinear pendulum in Simulink, and the generator voltage, current and power were simulated for different swing angles. The original design used a gear ratio of 13.3:1. Because of cost and time limits, the prototype was built with a ratio of 4.17:1 (50:12), and the simulation was repeated for this ratio. The simulation predicted peak generator voltages of 1.52 V to 7.5 V and peak power of 0.23 W to 5.6 W for swing angles of 20° to 120°.

The prototype was tested with a person seated on the swing. With no load connected, the boost module gave peak outputs of 0.981 V at 20°, 6.21 V at 45° and 8.17 V at 60°. The output rose with the swing angle, as the simulation predicted. The generator is driven on only one stroke of the swing, so the output appears as pulses. The gear ratio was found to have the largest effect on the output, and the final output is below the original 10 W design target. The project shows that a swing can generate electrical energy at small scale and could supply low-power loads, but the output still needs to be tested under load.

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1. Introduction

1.1 Background to the study

Energy is the ability to do work. It can be converted from one form to another such as heat, light, motion, chemical and electrical. In recent years, energy consumption has increased vastly due to population growth, economic development and industrialization, technological advancements, climate and environmental factors [2]. In South Africa, the leading contributor towards electricity generation is coal through the conversion of the chemical energy stored in coal. In the year 2024, coal was responsible for approximately 82% of the country's electricity generation [3]. Coal has the highest percentage of energy generated. However, the constant reliance on coal brings challenges such as long-term sustainability, poses health risks, pollution and rising costs due to market demand.

Despite the high energy production, South Africa still faces an energy crisis driven by aging coal infrastructure, mismanagement at Eskom, who is responsible for over 80% of electricity generation, and insufficient generation capacity [1].

This is where renewable energy stands out. Renewable energy is energy that never runs out and can be used for a wide variety of applications, such as solar, wind and hydro.

To help meet this demand, engineers found a solution by using renewable energy. Renewable energy is energy derived from natural sources that are naturally replenishing, such as wind, solar and kinetic energy. Unlike solar and wind conditions which depend on the weather to generate electricity, human-powered energy generation will occur while people use the equipment. In this project, it is proposed to generate electrical energy from the kinetic energy produced by the oscillatory motion of a playground swing that will be low cost, low-resource, low maintenance, safe for kids and environmentally friendly.

therefore, harvesting human-powered energy to produce electricity. This study investigates the design, simulation and implementation of the swing-based electrical power generator.

1.2 Objectives of this study

1.2.1 Problem Statement

The growing demand for electricity as well as the environmental and economic setbacks of fossil-fuel generation, has created a need for small-scale creative and innovative renewable energy solutions.

While large-scale solar and wind renewable energy is widely pursued, there still remains a significant gap in the development of low-cost energy harvesting systems that will be suitable for communities and educational applications. Kinetic energy produced during human activity in public spaces such as parks and playgrounds is part of a wide energy source that has been untouched. The pendulum-like motion that is generated by a person swinging contains mechanical energy which can be harvested and converted into electrical energy. This can serve as a practical and sustainable power source for low-power applications such as park lighting and phone charging station. Although the challenge lies in effectively converting the slow, bidirectional and irregular motion of a swing into smooth and usable electricity.

This study investigates the following key questions:

- How can the oscillatory kinetic energy of a swing be effectively converted to electrical energy?
- What mechanical design and configuration best optimises energy transfer from swing motion to the generator?
- Can a simulation model accurately predict the electrical energy generated from the system based on swing dynamics and operating conditions?
- Is the prototype capable of producing a sufficient electrical output for low-power application?

1.2.2 Purpose of the study

i. Energy access and community relevance

South Africa's national electricity access rate is approximately 84.4%, where rural areas have a much lower access rate of 75.3% [14]. This means that majority of the population, especially rural and underserved communities remain without reliable access to electricity. This is where this project of a proposed swing-based generator comes in, to serve as a small-scale energy harvesting system which offers as a practical and low-cost solution for supplying electricity to public spaces and off-grid communities that cannot be easily accessed by the conventional grid infrastructure.

ii. Environmental significance

The continue dominance of coal in South Africa's energy contributes significantly to greenhouse gases and air pollution, which brings about serious risks to public health and the environment. Energy harvesting devices can make use of renewable energy sources that are self-replenishing and abundant such as solar, wind, thermal and kinetic energy, which reduce the need for power transmission that cause energy losses and environmental impacts. Hy harnessing human kinetic energy, the swing generator being investigated in this study produces no harmful emissions and requires no fuel, making it an environmentally safe solution for electricity generation [15].

iii. Educational and social impact

Beyond the technical contribution, this study can be a major educational tool. Swings are enjoyed by people of all ages, making them an effective tool for demonstrating principles of energy conversion in a physical and engaging way. When the swing powers a light or a device in a park, it makes the concept of renewable energy visible for communities to observe and understand.

1.3 Scope and Limitations

Scope

This study focuses on the design, simulation and implementation of a small-scale swing-based electrical power generator that converts kinetic energy from the oscillatory motion of a playground swing into usable electrical energy. The scope of this study includes:

- The mechanical design of the energy conversion system.
- The electrical design of the output circuit.

- Development of the simulation model using Matlab/Simulink to predict the electrical output based on swing dynamics and operating conditions.
- Building a prototype of the swing generator.
- Evaluating the prototypes suitability for low-power applications such as powering a light.

Limitations

- Time constraints: The duration of this study is limited to one academic year which leaves less time for continuous testing and improvements.
- Budget constraints: The prototype is limited to low-cost components, which may not represent the best components that could be used in a real-life adaptation.
- Prototype scale: This study is for a small-scale lab model. Results may vary than a full-scale adaptation in a real playground.

1.4 Plan of Development

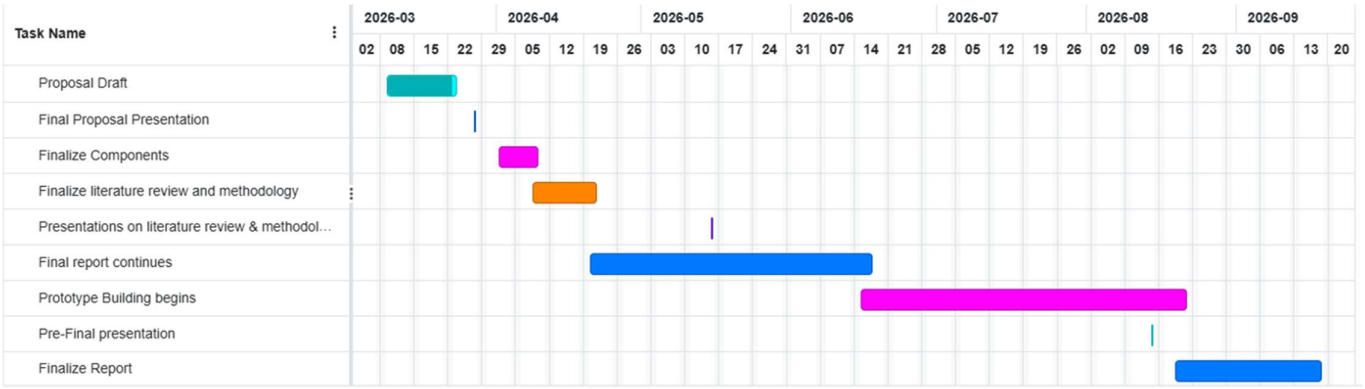


Figure 1: Gantt Chart

Summary of Gantt Chart:

- February, March: Proposal draft is written and proposal presentation.
- April, May: Finalising the literature review and methodology and finalise components as well as simulations. Presentation on literature review and methodology.
- April, May, June: Prototype building begins.
- September, October: Finalise report and pre-final presentation.
- October: Final project due.

2. Literature Review

2.1 Introduction to Renewable Energy Sources

Renewable energy is defined as the usable energy harvested from resources that are naturally replenishing such as solar, wind, hydro, geothermal heat, biomass [4]. Unlike fossil fuels such as coal, oil and natural gas, which accumulated after millions of years to form and are finite in supply, renewable energy sources are continuously replenished by nature and do not deplete over time [5]. An easily accessible source of renewable energy is a swing. Swings move in a pendulum-like motion, producing kinetic energy that can be transformed into electricity. Khan and Saeed et al. [6] indicates that it is possible to access this free energy source without using fossil fuels by placing energy harvesting systems on swings in parks and playgrounds.

Mullane et al. [7] study states that implementing swing-based kinetic energy systems demonstrates the usefulness of renewable energy in a hands-on manner. According to [8], powering lighting and small electronics through swings reduces the area's burden on the traditional grid. Swings are enjoyed by people of all ages, which makes them a great tool for teaching people about the importance of saving energy and protecting the environment. When a swing is used to power lights or small devices in a park, it can help communities become more aware of environmental issues and encourage communities to think of new and innovative ways to generate energy.

2.2 Swing Power Generations

Swing power generation refers to the process of converting kinetic energy from the swinging motion of a swing and converting it into electrical energy [9]. This concept has gained attention through its high potential for generating sustainable electricity from human activities, in playgrounds and public spaces to be specific. The idea is to capture the kinetic energy produced by the oscillatory motion and convert it into electrical power for low-power applications.

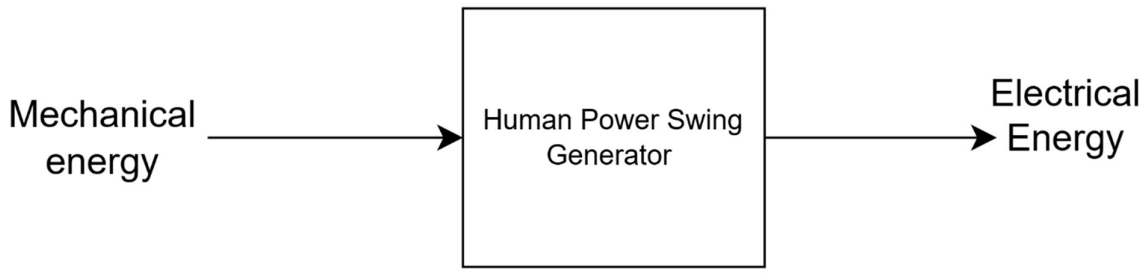


Figure 2: Simple block diagram showing conversion of energy

Riemer and Shapiro [10] explain the simple system behind power generation which involves attaching a generator or other energy harvesting devices to the joints of the swing and as the swing moves back and forth, the motion causes the generator to spin and produce electricity. Swing power systems can make use of different types of swings to capture energy.

- Simple playground swings with hanging seats can generate power [11].
- More advanced pendulum-style swing that uses counterweights and fuller arm motions [11].

Some designs use dual generators at the top and bottom pivots of the swing to harness energy in both directions of motion. The study of swing motion by Shastri and Bharath [12], investigates two different designs to drive a mechanical and electrical load:

- Harnessing bi-directional motion of the swing.
- Harnessing any one direction of the swing.

Although, some engineering challenges include optimising the generator design for varying swing speeds as well as building a durable system to withstand weather conditions and years of use. Mortley [13] describes how to build a swing set that generates electricity by using an old swing set by hanging a drive shaft below the top of the swing and attaching swings to it by roller clutches. As the swing begins motion, it would turn the shaft forward, and when swung backwards, the roller clutches will keep motion, and the momentum of the generator keeps forward motion. As the motions continues back and forth, the generator generates electricity.

2.3 Existing Research on Swinging Systems for Generating Energy

This section will discuss the studies that were done on swing systems.
Goku et al. [16] researched possible ways to reduce the high demand of power in India by developing a pendulum power generator, which will convert mechanical energy (kinetic) from a swinging pendulum weight to generate electricity. Ayneendra et al. [17] discussed using a swing to develop power in a system where mechanical energy is provided through a swing and is turned into electrical energy by a commutator and then stored within a battery. In order to transmit motion to the free wheel, which will rotate based on the swings angle of motion, the shaft will be combined to a sprocket. A chain will connect the sprocket and free wheel to convert angular motion into a whole rotation. The shaft that is attached to the free wheel will rotate the generator to produce the electricity.
Manjunatha et al. [18] developed a children's swing to play, that while in use it generates power. Anandhu, et.al. [19] had done a study which included a swing that uses sea waves to generate electricity as it is in use. The swing's simple structure causes the horizontal beam member to swing through an angle and continue to move back and forth.

3. Methodology

3.1 Research Approach

This study follows an experimental engineering approach, along with theoretical design, computer-based simulation and practical prototype construction to investigate the effect of harvesting kinetic energy from a swing.

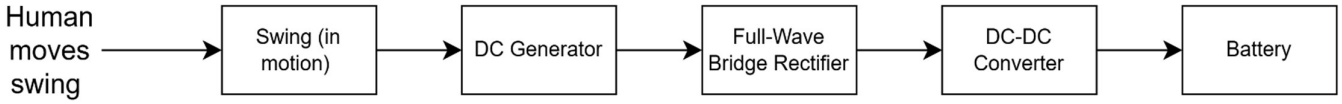


Figure 3: Block diagram showing the supposed system

Figure 3 shows the simplified block diagram of the proposed system for the swing power generator.

3.2 Specifications

After much research, it was determined that swings can be powered in various ways such as being human powered, driven by a motor and wind power.

Table 3-1: Comparison of Power supply:

Name:	Human Powered Swing:	Motor Driven swing:	Wind Powered swing:
Description:	A person sits on the swing and uses their body to make it move back and forth.	An electric motor is used to push the swing back and forth automatically.	Wind pushes the swing back and forth through a sail or paddle attached to the swing.
Advantage:	No fuel or electricity is needed. Anyone can use it and it teaches people about renewable energy in an interactive way.	The swing moves at a steady and controlled speed every time.	No person or motor is needed. It runs on its own as long as there is wind.
Disadvantages:	No power is produced when nobody is using the swing.	A power source is needed to run the motor, which means the system uses more electricity than it produces.	It only works when there is enough wind.
Why not chosen?	This method was chosen because it fits the criteria.	Using a motor to generate electricity would use	Wind is not always available.

		more electricity than it produces, therefore defeating its purpose.	

Therefore, after considering different types of ways to cause the swing to move, it has been finalised to use a human-powered source for this project to generate the kinetic energy in the swing.

3.3 Main components of this model:

- Swing frame
- Generator
- Free wheel(large)
- Generator Sprocket (small)
- Roller chain
- Full wave bridge rectifier
- DC-DC converter
- Rechargeable Battery
- Pillow block bearings
- Roller clutch bearings

3.3.1 Swing Frame:

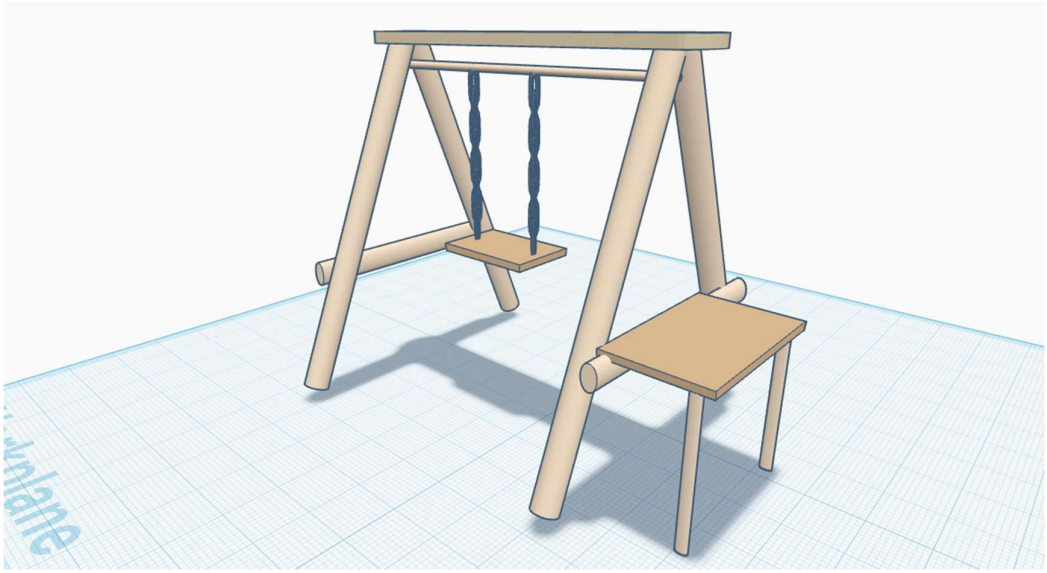


Figure 4: Simulation of swing frame

Table 3-2: Comparison of Swing Frame materials:

Name:	Metal frame:	Wooden Frame:	3D Printed Frame:
Description:	Made from aluminium that is welded or bolted together.	Made from wood that is joined together using screws or bolts.	Designed on a computer and printed using a 3D printer
Advantage:	Very strong and lasts a long time even under heavy use.	Simple to build using basic tools that is accessible.	Can be designed to exact measurements.
Cost:	Expensive.	Cheap.	Cheap.
Why not chosen:	Too heavy and expensive. Welding is not accessible.	This method was chosen.	Too weak for this prototype.

After comparing the three options of materials that can be used for the swing frame, the wooden frame has been finalised as the chosen method for this system prototype. It was chosen due to wood being cost-effective and construction made easier. Each leg beam will be 2100mm. The cross beam is 1700mm. The support beam at the top is 1500mm. The steel shaft is 1700mm.

3.3.2 Generator:

Table 3-3:Comparison of different options of Generators

Name:	Permanent Magnet DC Generator:	Stepper Motor Used Generator:	Brushless DC Generator:
Description:	Uses permanent magnets to produce electricity when the shaft is rotate.	Stepper motor that's repurposed to act as a generator by rotating its shaft to produce voltage.	Uses electronic commutation instead of brushes to produce electricity.
Advantage:	Simple to use.	Cheap and easy to find.	Very efficient and durable with no brushes to wear down.
Cost:	Low to moderate.	Very cheap.	Moderate to high.
Why not chosen:	This component was chosen.	Output voltage is irregular and difficult to control without	More expensive and requires additional electronic speed controllers.

		additional processing circuits.	

After considering three different types of generator options, it was concluded that the Permanent Magnet DC Generator will be utilised in this swing system. This generator was chosen due to easy implementation, affordable and best fitting for this system. The rating is 12V 320 RPM DC Motor.

3.3.3 Free wheel(large):

Table 3-4: Comparison of different material Freewheels

Name:	Semicircle Plywood Freewheel:	Metal Freewheel:
Description:	A semicircle shaped freewheel cut from plywood.	A metal freewheel that's used to convert back and forth motion into continuous rotation.
Advantage:	Cheap and easy	Strong, durable and smooth operation under load.
Cost:	Very cheap.	Moderate.
Why not chosen:	Too weak and prone to warping under repeated stress.	This component was chosen.

After considering both freewheel options, it was decided that the metal freewheel will be used. The semicircle plywood freewheel was considered due to the low cost and ease of acquiring but was not chosen since it was too weak for this system. The metal freewheel is a much more reliable solution. The freewheel will be 80 Teeth 25H.

3.3.4 Generator Sprocket (Small):

After considering two methods of combining the large freewheel and the generator, it was decided to make use of a sprocket and chain system. Connecting the pulley directly to the generator shaft was not chosen because it requires higher rotational speed than what the swing and freewheel will produce, therefore there will be insufficient RPM provided to generate electricity. The sprocket system can be designed to produce the needed RPM by choosing different sizes of the sprocket to either step up or step down the speed. The sprocket will be 7 Teeth 25H.

3.3.5 Roller chain

Table 3-5: Comparison of different types of roller chains

Name:	Rope:	Rubber Band Pulley:	Roller Chain:
Description:	A rope looped around two pulleys to transfer rotational motion.	A rubber band around two pulleys to transfer motion.	A metal chain that engages with sprocket teeth to transfer rotational motion.
Advantage:	Cheap and easy to source.	Very cheap and simple to set up.	No slipping.
Cost:	Cheap.	Cheap.	Moderate.
Why not chosen:	Prone to slipping and stretching under load.	Too weak, unable to handle the required load.	This component was chose.

3.3.6 DC-DC Converters:

Table 3-6: Comparison of different types of DC-DC Converters:

Name:	Buck Converter:	Buck-boost Converter:	Boost Converter:
Description:	Steps voltage down to a stable lower level.	Can both step voltage up and down depending on the input voltage.	Steps voltage up to a higher level than the input.
Advantage:	Very efficient at converting higher voltages down.	Handles variable input voltage in both directions.	Useful when the generator output voltage is too low and needs to be increased.
Cost:	Cheap.	Moderate.	Cheap.
Why not chosen:	Cannot handle input voltage below the target output since swing speed varies so voltage drops below 12V.	More expensive and unnecessarily complex since the generator voltage never exceeds 12V.	This method is chosen.

After evaluating all three voltage regulation options, it was decided that the boost converter would be used in the swing power generator system. The buck converter was ruled out as the swing moves at varying speeds causing the generator output voltage to frequently drop, which a buck converter cannot step up. The buck-boost converter was not used as it is more expensive and complex for this system since the generator is driven by a slow-moving swing it will never produce a voltage exceeding 12V. This system will make use of an XL6009 boost converter module.

3.3.7 Full Wave Rectifier Selection

For this project a single package rectifier module was selected as the most suitable rectifier for the electrical output circuit. The rectifier module has four diodes assembled inside one component, making it easy to connect directly to the generator output and the battery charging circuit. Since the swing produces motion in both directions, the rectifier module converts both directions of current into a single direction of DC current suitable for charging the battery. The module is cheap, widely available and reduces the risk of loose connections that could occur during the vibration caused by the swing motion. For this system, W10M Bridge Rectifier will be used.

3.3.8 Battery:

Table 3-7:Comparison of different types of Rechargeable Batteries:

Name:	Lead Acid Battery	Lithium Ion Battery	Nickel Metal Hydride Battery
Description:	Heavy rechargeable battery commonly used in cars	Lightweight rechargeable battery commonly used in phones	Rechargeable battery commonly used in household devices
Advantage:	Cheap and can handle being overcharged.	Lightweight and stores a large amount of energy.	Safe, reliable and widely available.
Cost:	Very cheap.	Expensive.	Moderate.
Why not chosen:	Too heavy and bulky for a small-scale prototype.	Too expensive for a small-scale prototype and requires a special charging circuit to prevent overcharging and damage.	This component was chosen.

For this system, I will be using a 12V 10Ah Sealed Lead Acid battery.

3.3.9 Pillow block bearings:

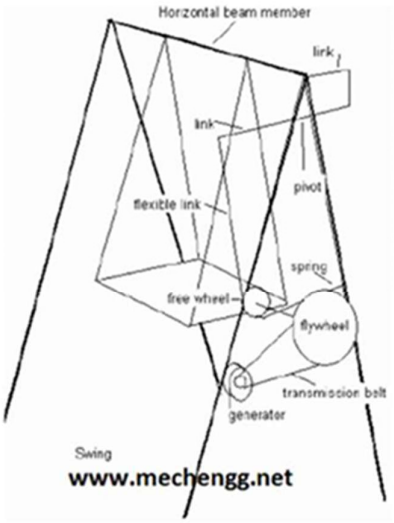
For this project two UCP210 pillow block bearings were selected as the most suitable bearing for supporting the horizontal shaft within the wooden frame. The pillow block bearing will be bolted directly onto the wooden frame, making installation simple. Since the shaft spans 1500mm between the two support points and carries a total load of 800N, the pillow block bearings provide the necessary load support at each end while allowing the shaft to rotate freely. The bearings can accommodate any slight misalignment that may occur during construction of the wooden frame. For this system two UCP210 pillow block bearings rated at 35kN will be used, one at each support point on the shaft.

3.3.10 Roller clutch bearings:

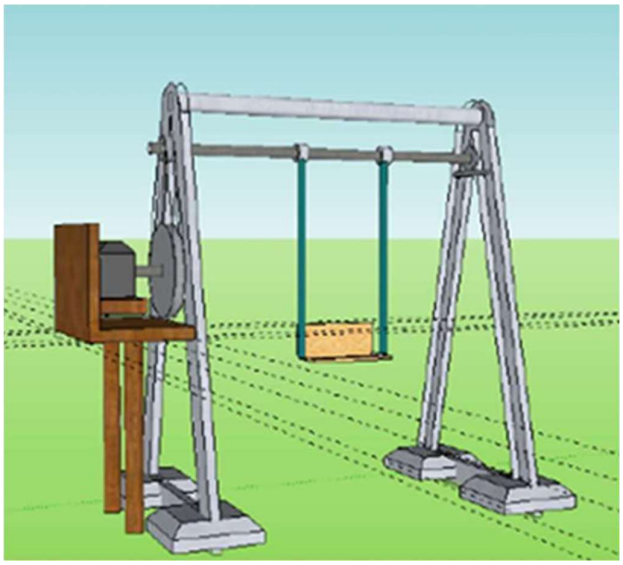
For this project two HFL5020 roller clutch bearings were selected as the most suitable bearing for connecting the swing to the horizontal shaft. The roller clutch bearing is a one way bearing that engages in one direction and freewheels in the opposite direction. When the swing moves in the forward direction the roller clutch bearing locks and transfers the motion to the shaft causing it to rotate, and when the swing returns in the opposite direction the bearing disengages and freewheels allowing the shaft to continue rotating in the same direction without being forced backwards. Two roller clutch bearings are used, one at each swing chain connection point on the shaft at positions B and C, ensuring that the load is evenly distributed across the shaft. For this system two HFL5020 roller clutch bearings with a 50mm bore diameter will be used.

3.4 Expected Design

Figure 5: Mechanical layout of the proposed swing generator



Swing Electricity Generation System Mechanical Project



3.5 Calculations:

Firstly, the output power should be determined in order to design the swing to meet the requirements.
For demonstration purposes, the swing is required to have enough output power for small appliances. This system is designed to supply power sufficient to turn an LED on, phone charging and small appliances.
A standard LED consumes 5-10W [20], a phone charging via a standard USB draws approximately 10W [21] and small appliances such as a fan consume around 10-25W. Based on these values, a realistic load of 10W is the finalised output power of this system. This value is achievable by a human-powered swing generator.

3.5.1 Torque on the vertical shaft:

Taking note of the forces acting on the vertical shaft and the torque:

Downward force:

$$T = F_1 \times L$$
$$T = M \times g \times l$$
$$T = (M + m) \times g$$
$$= (100 + 1.5) \times 9.81$$
$$= 995.715\ N$$

Where:
M- maximum weight sitting on the swing seat
m- weight of the swing seat
L- distance of the swing seat to the vertical shaft

Torque force during movement:

$$T = F_1 \times L$$
$$T = M \times g \times l$$
$$T = (M + m) \times g$$
$$= (100 + 1.5) \times 9.81 \times \sin(45°) \times L$$
$$= 101.5 \times 9.81 \times 0.7071 \times 1.1$$
$$= 774.485\ Nm$$
$$= 774\,500\ Nmm$$

Bending moment at B:

$$M_B = R_A \times 500$$
$$= 750 \times 500$$
$$= 375\,000 Nmm$$

Equivalent Torque:

$$T_e = \sqrt{M^2 + T^2}$$

$$= \sqrt{375\,000^2 + 774\,500^2}$$
$$= 860\,500\ Nmm$$

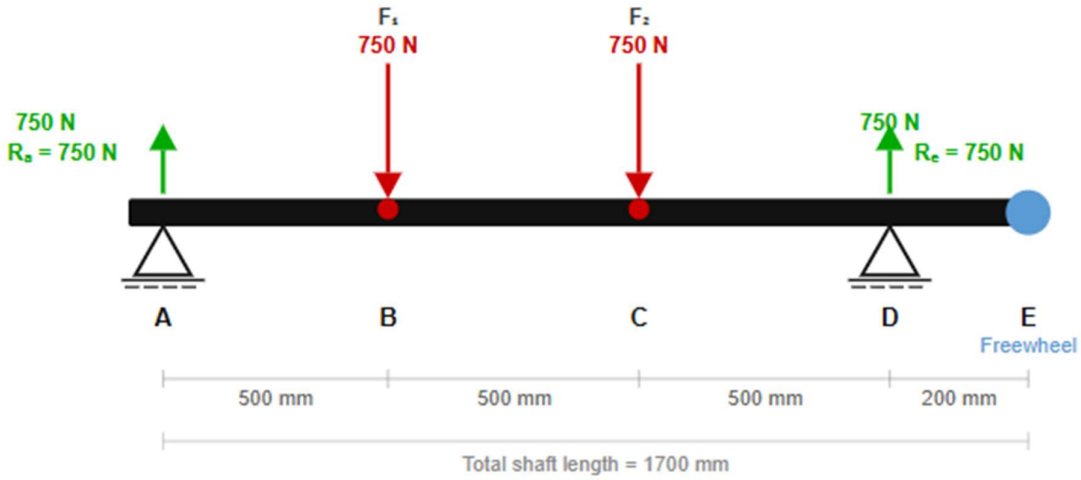


Figure 6: Diagram of forces acting on the support beam

Assume the weight of a person swinging (W)

$$W = M + m = 100 + 1.5 = 101.5\ kg$$
$$W = m \times g = (100 + 1.5) \times 9.81 = 799.515N = 800N$$

Allowing a dynamic factor of 1.5 for the swinging motion, the load is: $995.7 \times 1.5 = 1500\ N$
Therefore, load acting at B=1500/2= 750N
Load acting at C= 1500/2 = 750N

3.5.2 Shaft Diameter:

$$d^3 = \frac{32 \times \mathrm{Te}}{\pi \times 89.3}$$

$$d^3 = \frac{27\,536\,000}{280.530}$$

$$d = 46\ mm$$

Applying safety factor:

$$Safety\ factor = \frac{\sigma_{yield}}{\sigma_{allowable}}$$
$$= \frac{250}{89.3}$$
$$= 2.8$$

3.5.3 Generator rating:

To determine the rating of the generator:

$$f = \frac{1}{2\pi} \times \sqrt{\frac{g}{L}}$$
$$= \frac{1}{2\pi} \times \sqrt{\frac{9.81}{1.1}}$$
$$= 0.475\ Hz$$

Converting to RPM: $Swing\ RPM = 0.475Hz \times \frac{60sec}{min} = 28.5\ RPM$

Therefore, the swing completes 0.475 swings every second and in a full minute, it completes 28.5 full swings.

To find the required generator RPM, we need the generators voltage constant (Kv).

$$RPM = \frac{V_{output}}{K_v}$$

To be able to meet the requirement of powering a light, small appliance and charging cellphone, the output voltage $V_{output} = 12V$.

3.5.4 Sprocket sizing:

A driven sprocket needs at least 4-6 teeth to engage the chain without having faults or errors.

The required ratio can be determined by making use of the rated generated speed and the swing speed:

$$Ratio = \frac{N_{generator}}{N_{swing}}$$
$$= \frac{380}{28.5}$$
$$= Ratio = 13.3:1$$

Selecting freewheel and sprocket teeth:

The freewheel has 80 teeth and a driven sprocket of 6 teeth.
With this setup:

$$N_{generator} = 28.5 \times \left(\frac{80}{6}\right)$$
$$= 380\ RPM$$

This speed of 380 RPM will produce an output voltage of 14.25 V.
This exceeds the rated 12V, which is suitable for battery charging through the boost converter and charge controller.

3.5.5 Bearing selection:

Pillow Block bearing:
To find the dynamic load rating:

$$L10 = \left(\frac{C}{P}\right)^3 \times \left(\frac{10^6}{60n}\right)$$

Where:
L10- bearing life in hours
C- dynamic load rating
P-equivalent dynamic load
n- speed= 28.5 RPM

$$C = P \times \left(L10 \times \frac{60n}{10^6}\right)^{\frac{1}{3}}$$
$$= 750 \times 3.246$$
$$= 2.43\ kN$$

Where L10= 20 000 hours.

Roller Clutch bearing:

Torque capacity of each roller clutch bearing:

$$T_{each} = \frac{774.5}{2}$$
$$= 387.2\ Nm$$

3.5.6 Battery selection:

Table 3-8: Types of devices and their power ratings

Load device:	Power:
Phone charging	10W
LED light	10W
Small fan	10W

Assume worst case where one device is used for 5 hours:
$E = P \times t$
$= 10W \times 5$
$= 50Wh\ per\ day$

Capacity after Depth of Discharge (50%):

How much of the battery capacity is allowable to use before it needs recharging. Therefore, we will make use of the 50% rule to preserve battery life.

$$Ah = \frac{E}{V} = \frac{50}{12} = 4.17\ Ah$$

$$Ah_{rated} = \frac{4.17}{0.50}$$

$$= 8.33Ah$$

Now including 80% Efficiency:

$$Ah_{final} = \frac{8.33}{0.8} = 10Ah$$

Therefore, the 12V 10Ah Sealed Lead Acid battery is chosen.

3.6 List of finalised components:

- Swing frame (Consisting of 7 poles)
  - Leg poles: 4 Gum poles 2100mm, pole diameter of 75-100mm
  - Cross poles: 2 Gum poles 1700mm, pole diameter 75-100mm
  - Top support beam: 1 Gum pole 1500mm, pole diameter 75-100mm

- Horizontal Shaft (Rotating): Round bar 45mm * 1700mm
- Pillow block bearings: 2
- Roller clutch bearings: 45mm diameter, quantity=2
- Freewheel: 80 teeth
- Bicycle chain
- Generator sprocket: 6 teeth
- DC Generator: DC Motor 12V 320RPM
- Boost converter: XL6009 Boost Converter Module
- Rechargeable battery: 12V 10Ah Sealed Lead Acid

3.7 Simulations:

The non-linear pendulum was simulated using Simulink. The Non-linear Damped Pendulum consists of an object that swings back and forth.

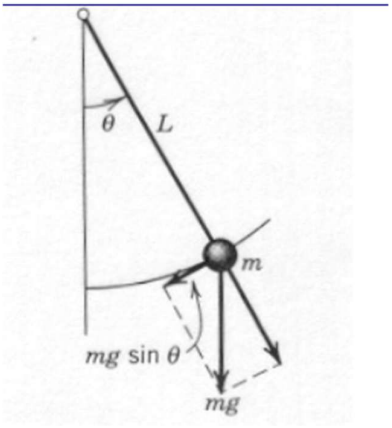


Figure 7: Diagram of a non-linear damped pendulum

A playground swing behaves similarly to a simple pendulum, where a mass **m** is attached to a rod of length **l**. As shown in figure 6, when the swing is displaces by angle θ from the vertical surface, the restoring force acting along it is:

$$F = -mgsin(\theta)$$

Applying Newtons second law of rotational motion:

$$mL^2\theta = -mgL\sin(\theta)$$

$$\theta = -\left(\frac{g}{L}\right) sin(\theta)$$

This is the nonlinear pendulum equation, where the sin(θ) shows that the angle between the vertical and the rod varies.

This equation was used in Simulink to simulate the motion of the swing movement.

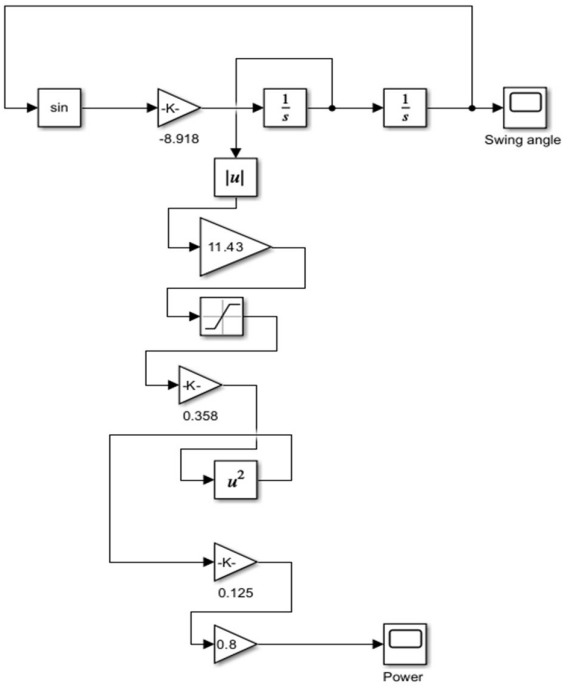


Figure 8: Simulation circuit of the swing mechanics

The Simulink model simulates the swing movement and the power output from the system. The model consists of two parts.

The first part models the swing as a nonlinear pendulum from the pendulum equation. The angular displacement θ is fed back through a sin block and multiplied by the gain -8.918, which represents -g/L. Two integrator blocks then produce the angular velocity θ̇ and angular displacement θ, which is fed back to close the loop. This feedback structure is what makes the model nonlinear.

The second part takes the angular velocity output and calculates the electrical power. The absolute value block handles the bidirectional swing motion due to the addition of a full wave rectifier, the ratio gain of 13.3 converts swing speed to generator speed, and the saturation

block clamps the generator at its rated speed of 33.51 rad/s to prevent the motor from going past the motors limit. The back-EMF constant of 0.358, the squaring block and motor resistance of 0.125 produce the electrical power, and a final efficiency gain of 0.8 accounts for system losses. The output is displayed on the Power scope in figure 9.

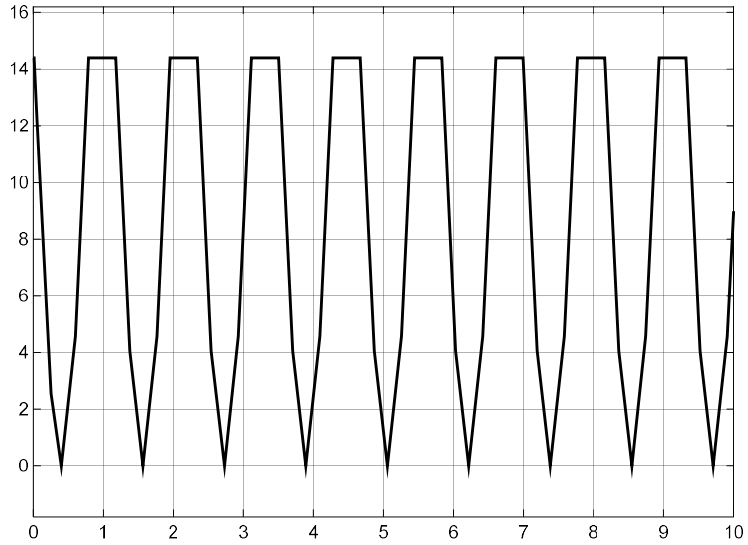


Figure 9: Plot of expected output power

Figure 9 shows the expected electrical output power of the generator over a 10 second simulation. Power is produced twice in every swing cycle, once on the forward stroke and once on the return stroke, because the absolute value block makes the generator speed positive in both directions. The power reaches its highest value when the swing passes through the lowest point of its path, where the angular velocity is greatest. It falls to zero at the two ends of each stroke, where the swing stops for a moment before changing direction. The flat tops of the waveform at about 14.4 W are caused by the saturation block, which limits the generator to its rated speed of 33.51 rad/s (320 RPM). At this speed the generator produces $0.358 \times 33.51 \approx 12$ V, which gives $12^2 \times 0.125 \times 0.8 \approx 14.4$ W. The power peaks are about 1.15 s apart, which is half the swing period. This shows that the power output follows the motion of the swing directly.

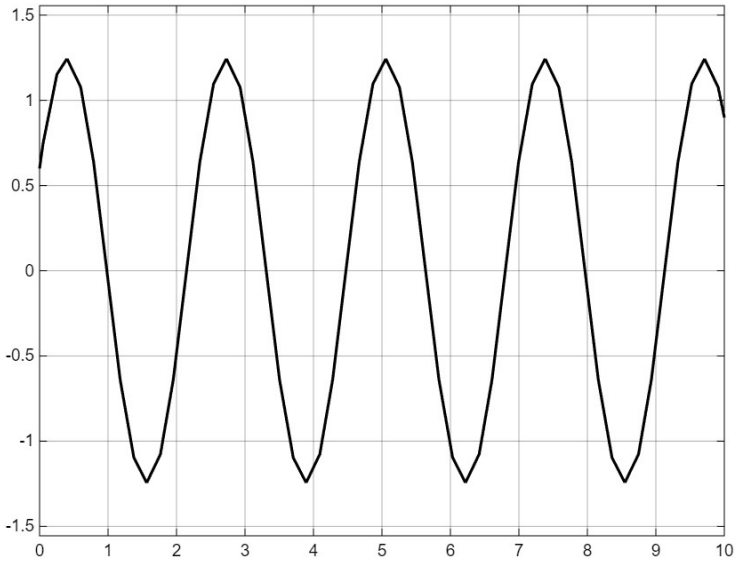


Figure 10: Plot of Swing angle

Figure 10 shows the angular displacement of the swing over the same 10 second simulation. The swing moves back and forth between about +1.24 rad and −1.24 rad, which is roughly ±71° from the vertical, with a period of about 2.33 s. This period is longer than the 2.10 s given by the small-angle formula for a 1.1 m pendulum. The difference comes from the sin(θ) term in the nonlinear pendulum equation: at large angles the restoring force is smaller than the linear approximation predicts, so the swing takes longer to complete each cycle. The amplitude stays the same throughout the simulation because the model contains no damping or electrical load on the swing motion. In practice, friction, air resistance and the load from the generator would slowly reduce the amplitude unless the person keeps swinging. Comparing Figure 10 with Figure 9, the power peaks line up with the points where the swing angle crosses zero, and the power falls to zero where the swing angle reaches its maximum.

3.7.1 Effect of Swing angle on Power output:

Table 3-9: Swing angle effect on output power

Swing Angle:	Initial condition (radian):	Peak power (W):
20°	0.35	2.3W
45°	0.79	12W
60°	1.05	20W
90°	1.57	40W
120°	2.09	57W

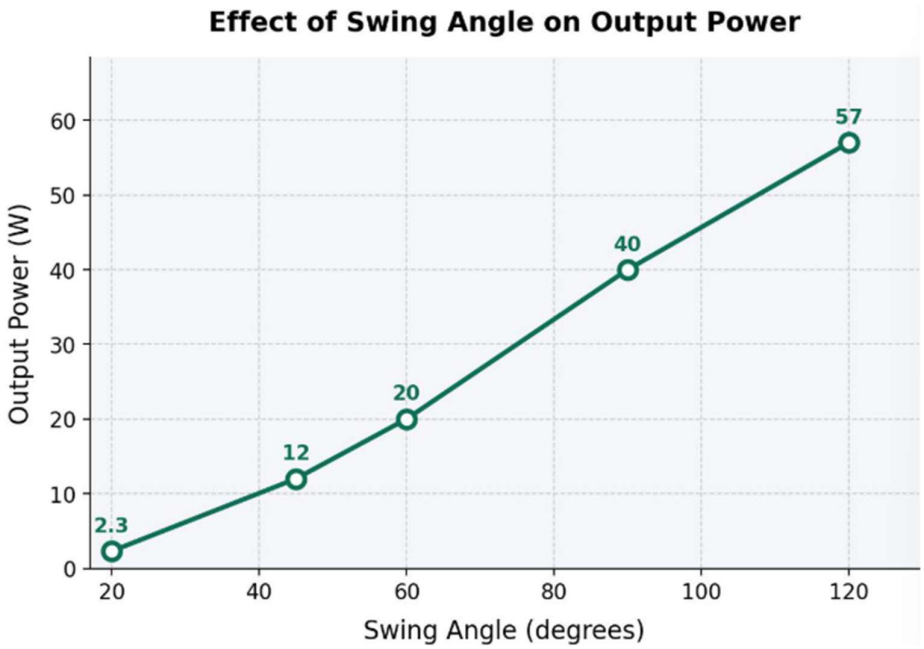


Figure 11: Plot of Output power VS swing angle

Figure 11 shows the peak output power for swing angles from 20° to 120°, using the values in Table 3-9. The peak power rises steadily with the swing angle, from 2.3 W at 20° to 12 W at 45°, 20 W at 60°, 40 W at 90° and 57 W at 120°. A larger swing angle means the swing falls through a greater height, so it reaches a higher speed at the bottom of each stroke and drives the generator faster. Because power depends on the square of the generated voltage, the power rises faster than the voltage as the angle increases. The 10 W design target is first reached at about 45°, so the user needs to swing at this angle or higher for the system to produce useful power.

3.7.2 Effect of Swing angle on Voltage:

Table 3-10: Voltage due to varying swing angle

Swing angle:	Peak voltage:
20°	4.8
45°	11
60°	14
90°	20
120°	24

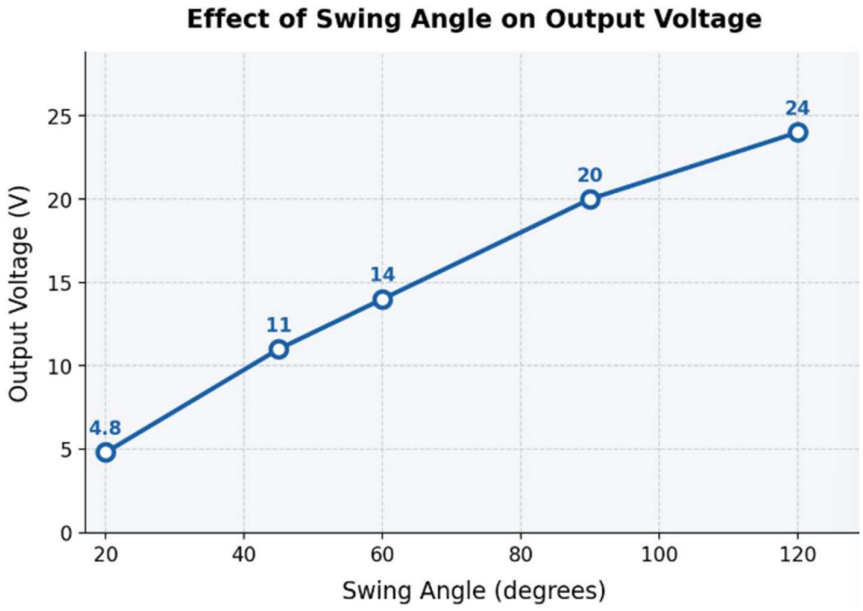


Figure 12: Plot of Output voltage VS Swing angle

Figure 12 shows the peak output voltage for the same range of swing angles, using the values in Table 3-10. The peak voltage rises from 4.8 V at 20° to 11 V at 45°, 14 V at 60°, 20 V at 90° and 24 V at 120°. The voltage of a PMDC generator is proportional to its shaft speed, so this curve follows the peak swing speed at the bottom of each stroke. The curve flattens slightly at larger angles, because the swing speed at the bottom grows with the square root of the drop in height, and the extra height gained per degree gets smaller as the angle approaches 90° and beyond. At 45° and above, the peak voltage is high enough to charge a 12 V battery or to feed the boost converter.

3.7.3 Effect of Swing angle on Current:

Table 3-11: Effect of varying swing angle on peak current

Swing angle:	Peak power (W):	Peak voltage:	Peak current:
20°	2.3W	4.8	0.48
45°	12W	11	1.09
60°	20W	14	1.43
90°	40W	20	2.00
120°	57W	24	2.38

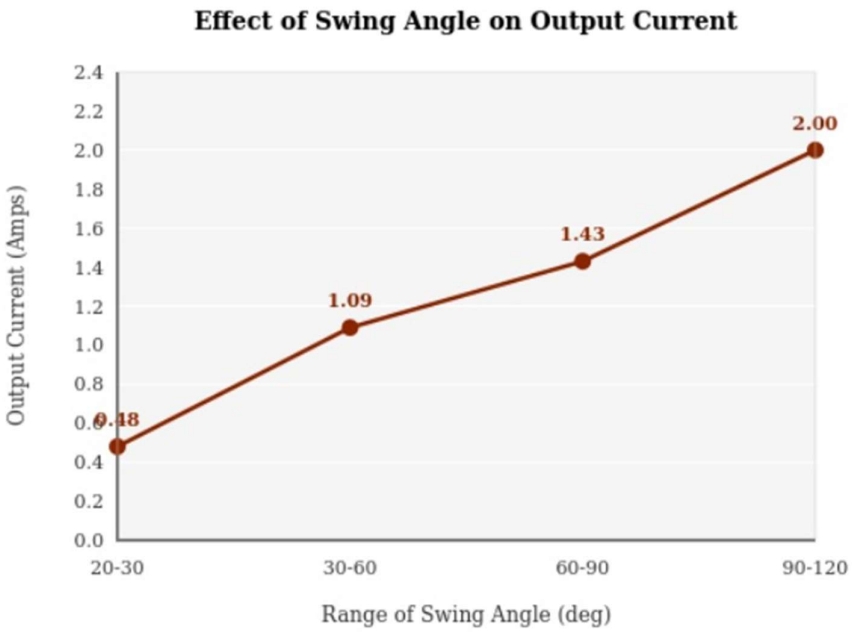


Figure 13: Plot of Output current VS Swing angle

From the simulation results, it can be seen that the swing angle has a direct effect on the output power, voltage and current. As the swing angle increases, the angular velocity at the bottom of each stroke increases, therefore driving the generator at a higher speed and producing greater power, voltage and current. At a low swing angle of 20°, the system produces only 2.3W, 4.8V and 0.48A, which is insufficient for any practical load. As the angle increases to 45°, the output improves significantly to 12W, 11V and 1.09A, meeting and exceeding the 10W design target. At 60° the system produces 20W, 14V and 1.43A, further confirming that the output continues to rise with increasing swing angle. These results demonstrate that the swing power generator is dependent on the effort and amplitude of the user's swing, and that an active swinging motion of at least 45° is necessary to generate sufficient electrical energy for practical low-power applications such as phone charging and LED lighting.

3.8 Design Modifications during Implementation:

During construction and testing of the prototype, two changes were made to the original design because of financial constraints, the availability of components and the time left for the project.

3.8.1 Replacing the roller clutch bearings with a bicycle freewheel:

The original design used two HFL5020 roller clutch bearings on the shaft to change the back and forth motion of the swing into rotation in one direction. This was changed to a bicycle freewheel fitted at the end of the shaft. The freewheel locks its sprocket when the shaft turns in the forward direction, and it spins freely when the shaft returns in the opposite direction. The chain between the freewheel sprocket and the generator sprocket therefore turns the generator in one direction only, which converts the bidirectional motion of the swing into unidirectional motion at the generator.

A bicycle freewheel was used because it is cheap, easy to find and does the work of the two roller clutch bearings with a single component. The generator is now only driven while the shaft turns in the driving direction.

3.8.2 Change in the gear ratio:

The original design used an 80 teeth freewheel and a 6 teeth generator sprocket, which gives a ratio of 13.3:1. This ratio could not be kept after the change to a bicycle freewheel, because a freewheel with a large number of teeth could not be sourced within the budget and time available. The first arrangement that was built used a 16 teeth freewheel and a 12 teeth generator sprocket, which gives a ratio of 16/12 = 1.33:1. With this arrangement the maximum voltage measured at the generator was only 1.19 V, which is too low to be useful.
To increase the ratio, a 50 teeth bicycle sprocket was welded onto the freewheel, with the help of the technicians and my classmate, so that the chain to the generator runs on the larger sprocket. The new ratio is 50/12 = 4.17:1, which is more than three times the previous ratio. For the same swing speed, the generator now turns more than three times faster.

Revised generator speed and voltage:

The swing speed of 28.5 RPM from Section 3.5.3 and the generator rating of 12 V at 320 RPM (0.0375 V per RPM) were used to calculate the expected output of each arrangement.
16:12 arrangement: Generator speed = 28.5 × (16/12) = 38 RPM, which gives 38 × 0.0375 = 1.43 V.
50:12 arrangement: Generator speed = 28.5 × (50/12) = 118.75 RPM, which gives 118.75 × 0.0375 = 4.45 V.
The calculated 1.43 V for the first arrangement is close to the measured 1.19 V. The measured value is about 84% of the calculated value, which is in line with the 80% efficiency used in the simulation and is caused by friction and chain losses. Applying the same factor to the 50:12 arrangement gives an expected output of about 3.7 V.
To meet the higher voltage target of 12V, I will make use of a DC Boost module.

3.8.3 Battery selection:

The battery that was meant to be storing power had been providing power to rotate the motor which was not the target for this prototype. It was decided that a powerbank will be used instead to store power and supply only when needed to.

3.8.4 Sizing of the shaft:

The 45 mm shaft calculated in Section 3.5.2 was machined down by about 5 mm, which gives a shaft diameter of about 40 mm. The end of the shaft was also machined down to 35 mm so that the bicycle freewheel could be fitted on it.

3.8.5 The effect of the new gear ratio on the output:

The simulation was first carried out for the arrangement that was built initially, which used a 16 teeth freewheel and a 12 teeth generator sprocket (ratio 1.33:1). The ratio gain in the Simulink model was set to 16/12, and the generator was made to turn in one direction only, as the freewheel does in the prototype. The model was run for swing angles of 20°, 45°, 60°, 90° and 120°.



Figure 14: Prototype setup with 16:12 gear ratio

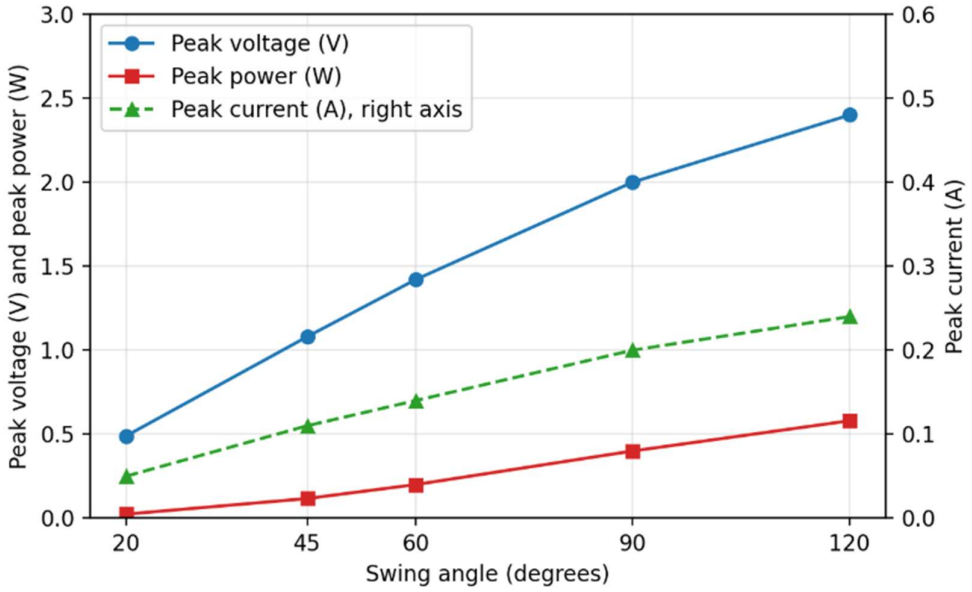


Figure 15: Output values for 16:12 gear setup

Table 3-12: Relationship of the swing angle on the output for 16:12 gear setup

Swing angle	Initial condition (radian)	Peak power (W)	Peak voltage (V)	Peak current (A)
20°	0.35	0.024	0.49	0.05
45°	0.79	0.12	1.08	0.11
60°	1.05	0.20	1.42	0.14
90°	1.57	0.40	2.0	0.20
120°	2.09	0.58	2.4	0.24

The peak power, voltage and current all increase with the swing angle, because a larger angle gives a higher swing speed at the bottom of each stroke. But the small ratio of 1.33:1 keeps the output very low. The peak voltage rises from 0.49 V at 20° to 2.4 V at 120°, and the peak power stays below 0.6 W at every angle. The output also falls to zero for about half of each swing cycle, because the generator is only driven on one stroke. The measured maximum voltage of 1.19 V lies between the simulated peaks of 1.08 V at 45° and 1.42 V at 60°, so the simulation is consistent with the measurement. This output was too small to be useful, which is why the ratio was increased by welding a 50 teeth sprocket onto the freewheel.



Figure 16: Setup of the improved 50:12 gear setup

The simulation was then repeated for the final arrangement, in which a 50 teeth sprocket was welded onto the freewheel and the generator sprocket has 12 teeth with a ratio 4.17:1. The ratio gain in the Simulink model was changed to 50/12, and the other settings were kept the same as for the 16:12 simulation. The model was run for swing angles of 20°, 45°, 60°, 90° and 120°. The table below gives the peak power, voltage and current for each angle and shows how they change with the swing angle. The output again increases with the swing angle, but it is much higher than for the 16:12 arrangement. The peak voltage rises from 1.52 V at 20° to 7.5 V at 120°, and the peak power rises from 0.23 W to 5.6 W. At 45°, for example, the peak voltage increases from 1.08 V to 3.4 V, and the peak power from 0.12 W to 1.15 W. This shows that increasing the gear ratio raises the generator speed and therefore the output. As before, the output falls to zero for about half of each swing cycle, because the generator is only driven on one stroke. Even with the larger sprocket the output remains below that of the original 13.3:1 design, since the ratio is still lower.

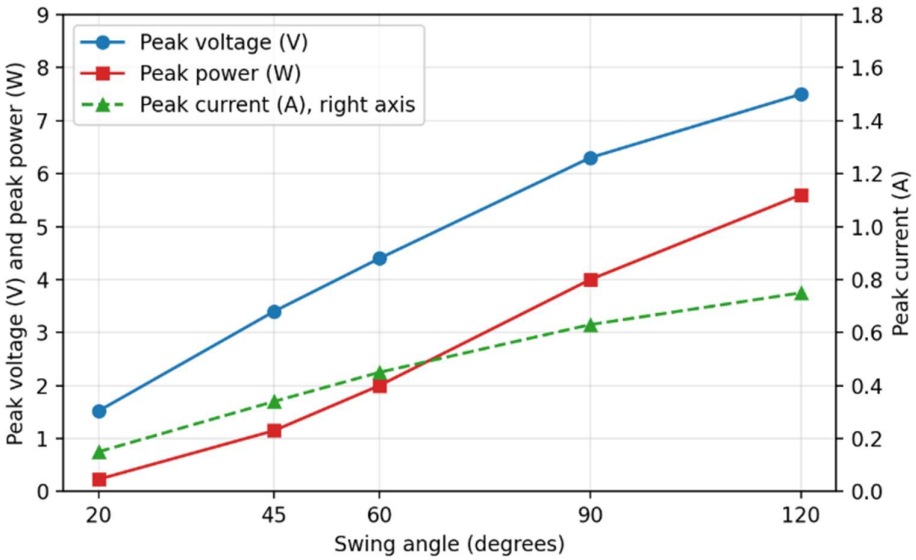


Figure 17: Output values for 50:12 gear setup

Table 3-13: Comparison of the original design and the final prototype (50:12)

Swing angle	Peak power, design (W)	Peak power, built (W)	Peak voltage, design (V)	Peak voltage, built (V)	Peak current, design (A)	Peak current, built (A)
20°	2.3	0.23	4.8	1.52	0.48	0.15
45°	12	1.15	11	3.4	1.09	0.34
60°	20	2.0	14	4.4	1.43	0.45
90°	40	4.0	20	6.3	2.00	0.63
120°	57	5.6	24	7.5	2.38	0.75

This table compares the simulation results of the original design with the built prototype. In both cases the peak power, voltage and current increase as the swing angle increases, because a larger swing angle gives a higher swing speed at the bottom of each stroke. The built prototype gives lower values at every angle. Its ratio is about 31% of the original, so the generator turns at about 31% of the speed and the peak voltage and peak current fall to about 31% of the design values. The peak power depends on the square of the voltage, so it falls to about 10% of the design values. At 45° the original design reaches 12 W, which meets the 10 W target, while the built prototype gives 1.15 W. Even at 120° the built prototype only reaches 5.6 W. The built prototype therefore shows that the swing can be converted into electrical energy, but at a lower output than the original design, which is the result of the smaller gear ratio that could be built within the cost and time limits.

4. Results

4.1 Test Setup:

The prototype was tested with the 50:12 gear arrangement. The swing was operated by hand and with a person seated and the output voltage was measured with a digital multimeter on the DC voltage range. Measurements were at the output of the DC boost module, which had no load connected. Because the generator voltage appears as short pulses on each driving stroke, the highest value displayed on the meter during each swing was recorded as the peak. A handheld meter updates slowly, so the true peak may be slightly higher. The figure below shows the prototype with a person seated on the swing during testing. The swing was released from different angles measured at 20°, 45° and 60°.



Figure 18: Prototype being tested with a person swinging on it

4.2 Measured Results:

The peak voltage was measured at the output of the DC boost module for swing angles of 20°, 45° and 60° and compared with the simulated generator voltage, which is the input to the module. The multimeter was used on the DC voltage range with no load connected, and the maximum value displayed during each swing was recorded as the peak voltage. The measured output rose with the swing angle, from 0.981 V at 20° to 6.21 V at 45° and 8.17 V at 60°. Larger, uncontrolled swings gave peak outputs of up to 16.7 V. On the reverse stroke, when the freewheel is disengaged, the readings were only in the millivolt range like 21.3 mV, which confirms that the generator is driven in one direction only. The measured output and the

simulated input are not the same quantity, but both rise with the swing angle. At 20° the simulated generator voltage is 1.52V, and the measured output stayed below 1 V, so the module did not boost. At 45° and 60° the simulated input is 3.4 V and 4.4 V, above that minimum, and the boost module raised the voltage to 6.21 V and 8.17 V. This agrees with the simulation result that a swing of about 45° or more is needed for a usable output. The output of the module is set by its adjustable trimmer, so these readings show the voltage level the circuit can reach and not the power delivered. With the earlier 16:12 arrangement the maximum generator voltage measured was 1.19 V, which lies between the simulated peaks of 1.08 V at 45° and 1.42 V at 60°.

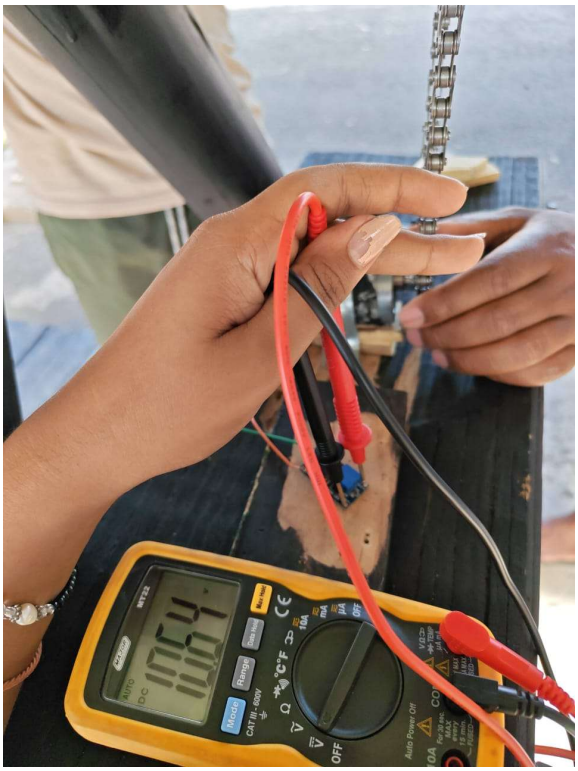


Figure 19: Output voltage measured using a digital multimeter

Table 4-1: Output voltage at different swing angles

Swing angle	Simulated generator voltage, 50:12 (V)	Measured boost module output peak (V)
20°	1.52	0.981
45°	3.4	6.21
60°	4.4	8.17

Large, uncontrolled swing (angle not measured)	-	16.7

The output increased steeply with the swing angle. The reading at 45° was more than six times the reading at 20°, and a further increase to 60° raised it again. The 20° reading stayed below 1 V, which means the boost module did not start at that angle. The voltages are the level set by the adjustable trimmer on the module, and they were taken with no load connected and with the swing operated by hand. They therefore show the voltage the circuit can reach, not the power it can deliver.

4.3 Comparison with the Calculations and Simulations:

The simulation predicts the voltage at the generator terminals, while the measurement was taken after the boost module, so the two cannot be compared value for value. They can be compared by trend. Both increase with the swing angle, and both show that a small release angle gives very little output.

The earlier 16:12 arrangement gives a direct comparison, because its generator voltage was measured. The hand calculation gave 1.43 V, the simulation gave 1.08 V at 45° and 1.42 V at 60°, and the measured maximum was 1.19 V, so the three results agree closely. For the 50:12 arrangement the hand calculation gives 4.45 V at the design swing speed, which is close to the simulated 4.4 V at 60°. The generator voltage of this arrangement was not measured separately, so it could only be compared through the boost module output.

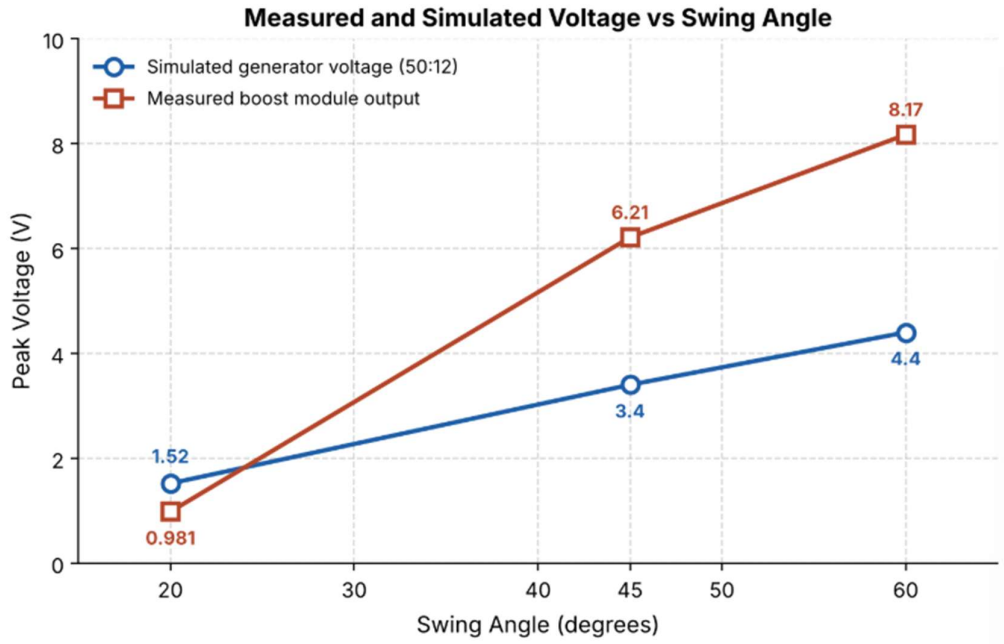


Figure 20: Measured VS Simulated Voltage Vs Swing angle

Figure 20 compares the measured boost module output with the simulated generator voltage for the 50:12 arrangement. Both rise with the swing angle, and the measured output sits above the simulated generator voltage at 45° and 60° because the boost module steps the voltage up. Dividing the measured output by the simulated input gives a step-up of about 6.21 ÷ 3.4 = 1.83

at 45° and 8.17 ÷ 4.4 = 1.86 at 60°. The step-up is almost the same at both angles, which is consistent with the simulation giving the correct trend in generator voltage.

4.4 Output on the Forward and Reverse Strokes

Table 4-2: Output on the forward and reverse strokes at 45 degrees

Stroke	Freewheel	Measured output
Forward (driving)	Engaged	6.21 V
Reverse (return)	Disengaged	21.3 mV

Table 4-2 compares the output on the two strokes of the swing. On the forward stroke the freewheel locks, the generator is driven and the boost module gives 6.21 V. On the return stroke the freewheel disengages, the generator stops turning and the output falls to 21.3 mV, which is about 0.3% of the forward-stroke value. This confirms that the freewheel converts the two-way motion of the swing into one-way rotation of the generator as intended. It also shows that power is only produced for about half of each swing cycle, so the output reaches the load as a series of pulses instead of a continuous supply.

4.5 Effect of the Gear Ratio on the Measured Output

Table 4-3: Comparison of the two gear ratios built

Arrangement	Gear ratio	Calculated generator voltage (V)	Measured result
16:12	1.33:1	1.43	1.19 V maximum at the generator; too low for the boost module
50:12	4.17:1	4.45	Boost module output of 6.21 V at 45° and 8.17 V at 60°

Table 4-3 compares the two gear arrangements that were built and tested. With the 16:12 arrangement the generator produced a maximum of only 1.19 V, which was too low for the boost module to work with, so the system gave no usable output. Welding the 50 teeth sprocket onto the freewheel raised the ratio more than three times, from 1.33:1 to 4.17:1. With this change the boost module produced 6.21 V at 45° and 8.17 V at 60°. This is the clearest measured result in the project, and it shows that the gear ratio decides whether the system can produce a usable output.

5. Discussion

5.1 Interpretation of the Results

The results show that the swing can be converted into electrical energy, but that the output of the final prototype is well below the original design target. The simulation predicted a peak generator voltage of 3.4 V at 45° and 4.4 V at 60° for the 50:12 arrangement, and the boost module raised the voltage to measured outputs of 6.21 V and 8.17 V. At 20° the simulated generator voltage was only 1.52 V and the module output stayed below 1 V, which shows that a small swing does not produce enough voltage for the circuit to work. The output therefore depends strongly on the swing amplitude, as the simulation predicted. For the final gear ratio the simulated peak power at 45° is 1.15 W, compared with 12 W for the original design, so the 10 W design target was not met. This is the result of the lower gear ratio of 4.17:1 instead of 13.3:1, which was the best that could be built within the cost and time limits as well as the availability of components.

5.2 Effect of the Design Changes

Replacing the roller clutch bearings with a bicycle freewheel allowed the swing to drive the generator in one direction at low cost, using a single component. The cost of this choice is that the generator is driven on only one stroke of the swing, which is discussed in Section 5.3. The change in gear ratio had the largest effect on the output. With the 16:12 arrangement the maximum generator voltage was only 1.19 V, which was too low to be useful. Adding the 50 teeth sprocket to the freewheel raised the ratio to 50:12 and increased the generator speed to more than three times its previous value, and the boost module then produced 6.21 V and 8.17 V at 45° and 60°. This shows that the gear ratio is the most important design parameter for this system, and that a ratio closer to the original 13.3:1 would give a higher output.

5.3 Limitations

The main limitation of the prototype is that the generator is only driven on one stroke of the swing. On the reverse stroke the freewheel disengages, the generator stops rotating and the output voltage drops to the millivolt range. The output therefore appears as a series of pulses with gaps between them, so the generator does not deliver continuous power, and the average output is lower than the peak values reported. This also makes it difficult to charge a battery or power bank steadily, because the boost module only receives an input voltage during the driving stroke.

The results have other limitations. The voltage was read from a handheld multimeter on a pulsed signal, so the true peak values may be slightly higher than those recorded. The swing was operated by hand, so the swing angles were approximate, and the generator voltage was not measured at the same time as the module output. Only 20°, 45° and 60° were tested, because larger angles were not practical on the prototype. The gear ratio of the final prototype (4.17:1) is also lower than the original design ratio (13.3:1), so the output is lower than the original design target of 10 W.

6. Conclusion

This project designed, simulated and built a swing-powered electrical generator that converts the back-and-forth motion of a swing into one-directional rotation using a bicycle freewheel and a chain drive. The swing's kinetic energy was converted into rotation of a permanent magnet DC generator, and a DC boost module was used to raise the low generator voltage. A freewheel made the rotation one-directional, and the gear ratio was found to be the most important design parameter. The original 13.3:1 ratio could not be built because of cost and time limits, and the final prototype used a 50:12 (4.17:1) ratio.
The mechanical structure was designed for safe use by a person. The swing shaft was sized using the equivalent torque method, which combines bending and torsion, with an impact factor of 1.5 for a 100 kg user. This gave a 45 mm EN8 shaft with a safety factor of about 2.8. During construction the shaft was machined down to about 40 mm, and its end was machined to 35 mm to fit the freewheel. The A-frame, pillow block bearings and chain drive held up during testing with a person seated on the swing, which shows that the structure can carry a real user while driving the generator.
The simulation predicted peak generator voltages of 1.52 V to 7.5 V and peak power of 0.23 W to 5.6 W for swing angles of 20° to 120°, which is below the 10 W design target. Testing measured a boost module output of 0.981 V at 20°, 6.21 V at 45° and 8.17 V at 60°, with output only on the driving stroke. The measured and simulated results rise with the swing angle in the same way, and the earlier 16:12 arrangement agreed reasonably well with its hand calculation and simulation. The prototype therefore shows that the concept works at small scale, but its output is lower than the original design because of the smaller gear ratio, and the generator is only driven on one stroke of the swing.

Looking back at the research questions in Section 1.2, the first question asked how the oscillating motion of a swing can be converted into electrical energy. The project showed that a one-way mechanism, a chain drive that raises the speed, and a PMDC generator followed by a boost module do this successfully, since a clear voltage was produced on every driving stroke. The second question asked which mechanical configuration best transfers energy from the swing to the generator. Here the results showed that the gear ratio matters more than any other design choice. Raising the ratio from 16:12 to 50:12 increased the generator speed more than three times and turned an output of 1.19 V, which was too low to use, into one the boost module could work with. The third question asked whether a simulation can predict the output of the system. The nonlinear pendulum model in Simulink did this well: for the 16:12 arrangement the hand calculation gave 1.43 V, the simulation gave 1.08 V to 1.42 V for 45° to 60°, and the measured value was 1.19 V. The model also correctly predicted that a swing of about 45° or more is needed for a usable output, which the boost module confirmed by only starting above 20°. The fourth question asked whether the prototype can produce enough output for low-power applications. This could only be partly answered. The boost module reached 6.21 V at 45°, 8.17 V at 60°, and up to 16.7 V during larger swings, but the 10 W target was not met with the built gear ratio, and no load test was done to confirm charging.

Overall, the project met its aim of showing that the kinetic energy of a swing can be harvested and converted into electrical energy using simple, low-cost components that are available locally. It also showed that the simulation can be trusted as a design tool for this kind of system, since it matched the measured behaviour once the as-built gear ratio was used. The main gap between the design and the prototype came from the components that could be sourced within the budget and time available, not from the concept itself. With a gear ratio closer to the original 13.3:1 and a way to drive the generator on both strokes, as set out in Chapter 7, the simulation shows that the system can exceed 10 W at a swing angle of 45°. That would make it practical for low-power uses such as phone charging and LED lighting in parks and off-grid communities.

7. Recommendations

Based on the results of this project, several improvements are recommended for the swing power generator. The gear ratio should be increased first, because the final ratio of 4.17:1 is much lower than the original design ratio of 13.3:1, and this was the main reason the output stayed below the 10 W target.
A larger freewheel or a two-stage chain drive would bring the ratio closer to 13.3:1. Larger sprockets should be attached with bolts or fixed to a separate collar instead of being welded directly onto the freewheel, because the heat can damage the freewheel mechanism.

The generator is also only driven on one stroke of the swing, so a second freewheel and generator, or a mechanism that converts both strokes into one-directional rotation, would use the full motion of the swing. A flywheel on the generator shaft or a larger capacitor on the generator output would keep the generator turning and reduce the gaps between voltage pulses, and a generator with a lower rated speed or a higher voltage constant would produce a higher voltage at the slow speed of the swing.
And lastly, adding another swing seat with different swing directions will keep the generator in constant motion.

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9. EBE Faculty: Assessment of Ethics in Research Projects

UNIVERSITY OF ZULULAND HUMAN RESEARCH ETHICS COMMITTEE
(Reg No: REC-171110-030)

UNIVERSITY OF ZULULAND

ETHICS APPROVAL LETTER

Certificate Number	**REC-171110-030 PGH 2026/355**								
Project Title	**Swing Electrical Power Generator**								
Principal Researcher/ Investigator	R, Moodley								
Supervisor and Co-supervisor	Dr Franck C. Mushid								
Department	Electrical Engineering								
Faculty	Science ,Agriculture and Engineering								
Type of Risk	Low Risk – Desktop, fieldwork, or laboratory								
Nature of Project	**Honours/4th Year**	**X**	Master's		Doctoral		Departmental		

The University of Zululand's Human Research Ethics Committee (HREC) hereby gives ethical approval in respect of the undertakings contained in the above-mentioned project. The Researcher may therefore commence with data collection from the date of this Certificate, using the certificate number indicated above.

SPECIAL CONDITIONS:

(1) This ethics approval letter is valid for 1 year from the date of issue.
(2) Principal researcher must provide an annual report to the HREC in the prescribed format [due date- 14 September 2027]
(3) The HREC must be informed immediately of any material change in the conditions or undertakings mentioned in the documents that were presented to the meeting.
(4) Under the Protection of Personal Information Act, 04 of 2013 ("POPIA"), researchers have a general legal duty to protect the information they process. They must ensure the security and protection of any personal information processed through the research and provide a compliant and consistent approach to data protection. The information collected via interviews must be for research purposes only. No personal information such as opinions, views, and academic background may be linked to the respondents' identity or shared with anyone for marketing purposes or otherwise.

The HREC wishes the researcher well in conducting research.

Naidoo

Prof. Kanagie Naidoo
Chairperson: Human Research Ethics Committee
14 September 2026

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